Internship PROJECT

On

Student Records using CRUD

Computer Science & ENGINEERING

SUBMITTED BY

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NEW GOVERNMENT POLYTECHNIC PATNA – 13

[A constituent institute of sbte, bihar] science, technology, & technical education department

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PROJECTS'S DECLARATION

I hereby submit my internship project , which was completed during our summer vacation using Python .

This project is a lightweight and efficient **Student Management System** developed using the Python Flask framework. It is designed to manage student records effectively, allowing users to perform essential **CRUD** (Create, Read, Update, Delete) operations on student data. The core functionality revolves around handling student information — such as name, roll number, and timestamp of registration — stored in a local JSON file.

This approach eliminates the need for a complex database, making it ideal for small institutions, academic demonstrations, or learning purposes.

The application begins with a homepage that redirects users to a list of all registered students. Users can then add new students via a form, where they input the student's name and roll number. A key feature of this system is its validation mechanism to prevent duplicate roll numbers. Each time a new student is added, the system records the current timestamp using the **pytz** module to ensure accurate time tracking based on **the Asia/Kolkata timezone.**

Users can also edit existing student information or delete records, with all changes being immediately saved in the students.json file. The data management logic is handled through helper functions like load_students() and save_students(), which abstract the JSON read/write operations to maintain clean and readable code. The use of Flask's routing system and the Jinja2 templating engine ensures a responsive and clean UI/UX experience.

The project structure is minimal and easy to navigate, making it ideal for beginners in web development. It avoids the complexity of using a relational database like **MySQL or PostgreSQL ,** yet demonstrates real-world web application patterns including form handling, data validation, redirection, and dynamic rendering.

Furthermore, this project serves as an excellent base for future extensions. It can be scaled by integrating an SQL-based backend, implementing user authentication for access control, or even deploying the system online using platforms like **Heroku or Render.**

PROJECT'S DETAILS

Student Management System using Python and Flask

This Student Management System is a simple yet functional web-based application developed using Python with the Flask micro-framework. It is designed to help manage student records efficiently in a minimal environment, using JSON files as the backend for data storage. The project enables users to perform CRUD (Create, Read, Update, Delete) operations on student data through an interactive web interface**.**

The system maintains a list of students where each student record includes a roll number, name, and the date and time of registration. All data is stored in a JSON file (students.json), and utility functions such as load_students() and save_students() handle the reading and writing of this file. These operations ensure that all updates to student records are saved persistently.

The Flask framework handles routing and logic for various functionalities:

- **/add** – to add a new student
- **/edit/<roll>** – to edit a student's data
- **/delete/<roll>** – to delete a student record
- **/students** – to view all student records

The application uses HTML templates with Jinja2 for rendering dynamic content. Timezones are managed using the pytz module to ensure correct date and time logging (specifically for Asia/Kolkata timezone). Form validation prevents duplicate roll numbers from being added.

The frontend is minimal and clean, focusing on ease of use for users such as school staff or administrators. Since the application does not rely on SQL or complex databases, it is lightweight and easy to deploy locally or on small-scale hosting platforms. It is especially suitable for use in academic institutions, small coaching centers, or as a beginner's project to understand web development with Flask.

Key Features:

- Simple web-based interface
- CRUD operations on student data
- JSON-based persistent storage
- Timestamp logging with timezone support
- Duplicate entry prevention
- Clean UI using HTML templates

Tools and Technologies Used

1. Python

Python is a high-level, interpreted programming language known for its readability and simplicity. It is widely used in web development, automation, data analysis, and artificial intelligence. In this project, Python is the core language used to build the application's logic and handle backend operations.

2. Flask

Flask is a lightweight and flexible micro web framework for Python. It allows developers to build web applications quickly with minimal code. Flask handles routing, request handling, and rendering of templates. It is suitable for small-scale projects and ideal for learning purposes.

3. JSON (JavaScript Object Notation)

JSON is a lightweight data-interchange format that is easy for humans to read and write. In this project, JSON is used as the data storage system instead of a traditional database. Student data is saved and loaded from a **`students.json`** file, making the application simple and file-based.

4. HTML & Jinja2 Templates

HTML (HyperText Markup Language) is used to design the front-end interface of the application. Flask uses Jinja2 as its templating engine, allowing Python variables and logic to be embedded directly in HTML files. This makes it possible to render dynamic content based on backend data.

5. pytz

The **`pytz`** library is used to handle time zones in Python. In this project, it ensures that the timestamp saved during student registration is based on the "Asia/Kolkata" timezone, which is important for consistency in record-keeping .

6. Bootstrap (Optional)

Bootstrap is a front-end CSS framework used for styling HTML components to make the user interface more responsive and visually appealing. Though optional, it can be used in this project to enhance the look and feel of the pages .

FEEDBACK OF OUR PROJECT

The **Student Management System** developed using Python and Flask is a well-structured and efficient solution for managing student records in a simplified manner. The project showcases a clear understanding of web development fundamentals, particularly in using Flask for backend logic and JSON for data storage. It effectively demonstrates the use of routing, form handling, and data persistence—core elements of any web application.

The application is lightweight and user-friendly, with a clean interface that enables users to easily add, update, view, and delete student records. The inclusion of validation for duplicate roll numbers and the use of timestamps (with timezone handling) shows attention to detail and practical implementation of real-world requirements.

One of the key strengths of this project is its simplicity and ease of deployment, as it does not rely on any external database. This makes it ideal for small educational institutions, internal use in classrooms, or as a beginner-level demonstration of CRUD operations in Flask.

However, there are a few areas that can be improved or extended in the future. Adding user authentication, exporting records to CSV, or switching from JSON to an SQL database like SQLite or PostgreSQL could make the system more scalable and secure. Additionally, improving the UI using CSS frameworks like Bootstrap would enhance the overall user experience.

Overall, this project is a strong demonstration of backend development skills and project planning. It successfully meets its objectives and lays a solid foundation for future enhancements.